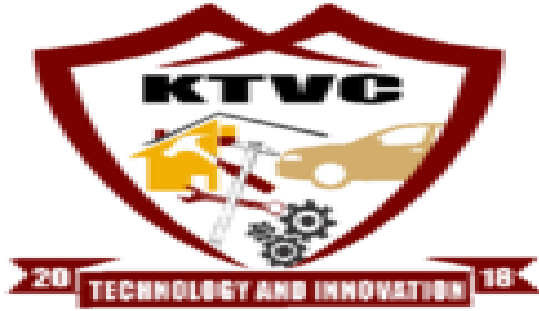




KIPKABUS TECHNICAL AND VOCATIONAL COLLEGE

STACTURED PROGRAMMING

DICT 23M MOD 2

END OF TERM

MARCH 2024

TIME: 2 HOURS

Instructions to the candidates

- 1. NO PHONE ARE ALLOWED**
2. The paper consist of **EIGHT** questions
3. Answer any **SEVEN** questions
4. No books or any written material are allowed in an examination room
5. A candidate who breaks examination regulations, or commit misconduct, will be disqualified from the examination room

1	2	3	4	5	6	7	8	TOTAL

1.
 - a) Define the following
 1. Computer program (2mks)
 2. Programming (2mks)
 3. Compiler (2mks)
 - b) Outline four functions of a compiler (4mks)
2.
 - a) Write a Pascal program code to display "hello world" (4marks)
 - b) State and explain data types in C programming (6marks)
3. Explain the functions of any five operators used in C programming with an example use in each case (10marks)
4. Inspect the code below and answer the subsequent questions

```
#include <stdio.h>
```

```
int main() {
    int marks;

    // Input the student's marks
    printf("Enter the student's marks: ");
    scanf("%d", &marks);

    // Check the grade based on marks
    if (marks >= 90) {
        printf("Grade: **A**\n");
    } else if (marks >= 80) {
        printf("Grade: **B**\n");
    } else if (marks >= 70) {
        printf("Grade: **C**\n");
    } else if (marks >= 60) {
        printf("Grade: **D**\n");
    } else {
        printf("Grade: **F**\n");
    }
}
```

```
return 0;
```

```
}
```

- a) Draw a program flow chart to represent the program in (4) above (8marks)
- b) What is the meaning of the term Source program (2marks)

5.

a) Inspect the Pascal program below and rewrite it correctly

(6marks)

```
program CalculateTriangleArea;

var
    Base Height,Area: real

begin
    { Input the base and height of the triangle }
    Enter the base of the triangle: ;
    readln(Base);
    writeln('Enter the height of the triangle: ');
    readln(H);

    { Calculate the area of the triangle }
    Area := (Base Height) / 2

    { Output the area
    writeln('The area of the triangle is: ', Area);
end.
```

b) Explain any Two functions of an assembler

(4marks)

6.

a) What is the difference between a compiler and an assembler

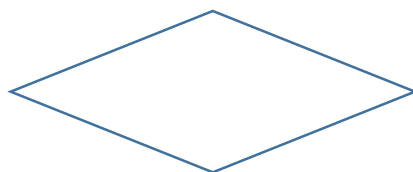
(2marks)

b) State the features of low-level languages

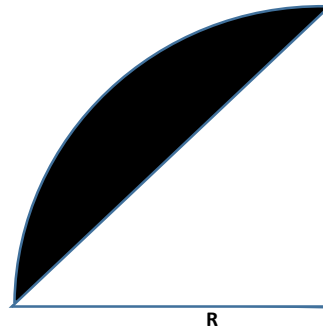
(4marks)

c) What is the function of this shape when drawing flowcharts

(2marks)



7. The figure below shows a piece of land. Write a C program to find the area of the shaded region. The distance R is the radius of the circle. (10 marks)



- 8.
- a) What is the meaning of the term global variable (2 marks)
 - b) Distinguish between the selection and quick sort method (4 marks)
 - c) Outline the use of the following in C programming language (3 marks)
 - I. `{ }`
 - II. `[ ]`
 - III. `#`
 - d) What was the function of COBOL third generation language (1 mark)